

① Image enhancement in Spatial domain

Process of Improving visual quality of an image so it becomes easier for humans or computers to analyze. In spatial domain enhancement is performed directly modifying pixel value of the image.

$$g(x, y) = T[F(x, y)]$$

* Grey level transformation :- Intensity values of individual pixels to improve brightness and contrast.

$$s = L - 1 - r$$

- log transformations :- $s = c \log(1+r)$

- power law transformations :- $s = cr^{\gamma}$

- Contrast stretching :- $s = \frac{(r - r_{\min})}{r_{\max} - r_{\min}} \times 255$

- Histogram Equalization :- Intensity values uniformly across the entire gray level range.

* Spatial Filtering :- Enhances an image by applying a mask level neighboring pixels.

Two major categories. :-

① mean filter :- replaces each pixel by average of neighbour pixels.

② median filter :- center pixel with the median of neighbour values.

Image Smoothing vs Image Sharpening

→ Image Smoothing & Image Sharpening are in the domain image enhancement techniques.

* Image smoothing reduces unwanted noise and removes by averaging or replacing neighbour pixel values.

* Image sharpening is identifying edges and lines.

Feature	Image Smoothing	Image Sharpening
Purpose.	Reduce noise & remove unwanted objects.	Enhance boundaries and details.
Effect	make image smoother	make image clearer
Frequency Components	reduce high frequency components.	Enhance high-frequency components.
Main Filters.	Mean, Gaussian, median	Laplacian, Sobel
used when	Image contains noise.	Image is blurred.

Image Enhancement in Frequency Domain.

Frequency domain Enhancement modifies Frequency Components instead of pixel values.

$$G(u,v) = H(u,v) F(u,v)$$

- Steps :-
- (i) read image
 - (ii) apply fourier transform
 - (iii) apply filter
 - (iv) apply inverse fourier transform
 - (v) obtained enhance image

Low pass Filter

Working :-

- * Passes low frequencies
- * Block high frequencies

Effects :-

- * remove noise
- * produce smooth images
- * Blur edges

Applications :-

- * medical image
- * satellite image

High pass Filter

Working :-

- * Pass high frequencies
- * removes low frequencies

Effects :-

- * sharpens image
- * Enhances image

Applications :-

- * edge detection
- * finger print recognition

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Image segmentation

→ Process of dividing an image into meaningful regions or objects. and makes the image clarity.

Techniques :-

① Point detection :- Detects isolated pixels using markers.

⊗ Advantages are its simple & fast.

⊗ Limitations :- sensitive to noise

⊗ Applications :- road pixel detection.

⊗ Line detection :- Detects horizontal, vertical and diagonal lines using convolution masks.

— detects straight line.

— Not suitable for curved objects.

— road detection & document analysis.

② Thresholding :- separates foreground & background using intensity values.

$$g(x,y) = \begin{cases} 1, & f(x,y) \geq T \\ 0, & f(x,y) < T \end{cases}$$

Edges represents sudden changes in intensity and usually correspond to object boundaries. Edge detection, edge linking and boundary detection are essential for object recognition & detect.

(i) Edge detection :- rapid intensity changes and usually correspond to object boundaries

(ii) Edge linking :- connects broken edge pixels into continuous curves using.

- Hough transforming
- contour following.

~~(iii) Boundary detection :- identifies the complete outline of segmented objects using edge information~~

- shape analysis.
- object counting
- defect detection.